Handoff Guide

Start here

What this is

UniQOS DB is an embedded contacts/messages/calls database for the STM32H723 (Cortex-M7, FreeRTOS), designed to manage up to **10,000 contacts** and roughly twice as many messages on an SD card, with power-fail-safe writes and a RAM footprint small enough to share the chip with the rest of the UniQOS phone firmware. The database, its on-hardware demos and unit tests, two benchmark suites, and a set of offline Python design studies all live in this one repository.

This `docs/` set is a **handoff guide**: it orients a new contributor and links out to the much more detailed material that already exists in the repository (`README.md` and the 29 files under `notes/`) rather than repeating it. Read this set first, then go deeper via the pointers in each chapter.

> **Already in the repo and worth knowing about before you read further:**
> - `README.md` – the original architecture write-up (relational model, SD layout, hash table reconstruction, rollback journal, tombstoning), with diagrams.
> - `notes/` – 29 markdown files: design rationale, debugging postmortems, build options, demo/CLI usage, and one file per GTest suite under `notes/unit_tests/`.
> - `diagrams/UniQOS DB Architecture.drawio` – 25 pages covering the DB's internal data structures and flows in detail. This handoff set's own diagrams (`docs/diagrams/handoff-overview.drawio`) are a separate, smaller set covering what that file doesn't: the repo layout, the build option matrix, the benchmark pipeline, and current project status.

How to read this set

01 – Architecture Overview System layers and the module responsibility table for `source/app`. Read this first if you're new to the codebase.

02 – Build, Test & Flash How to configure, build, test and flash both the host and STM32 targets, and how to read test results off the board.

03 – Benchmarking & Design Evidence How the hashing and collision design decisions were validated, from the offline Python studies through to the on-target benchmarks.

04 – Project Status & Next Steps What's stable, what's currently mid-flight in the working tree, and the known open items worth picking up next.

Repo map

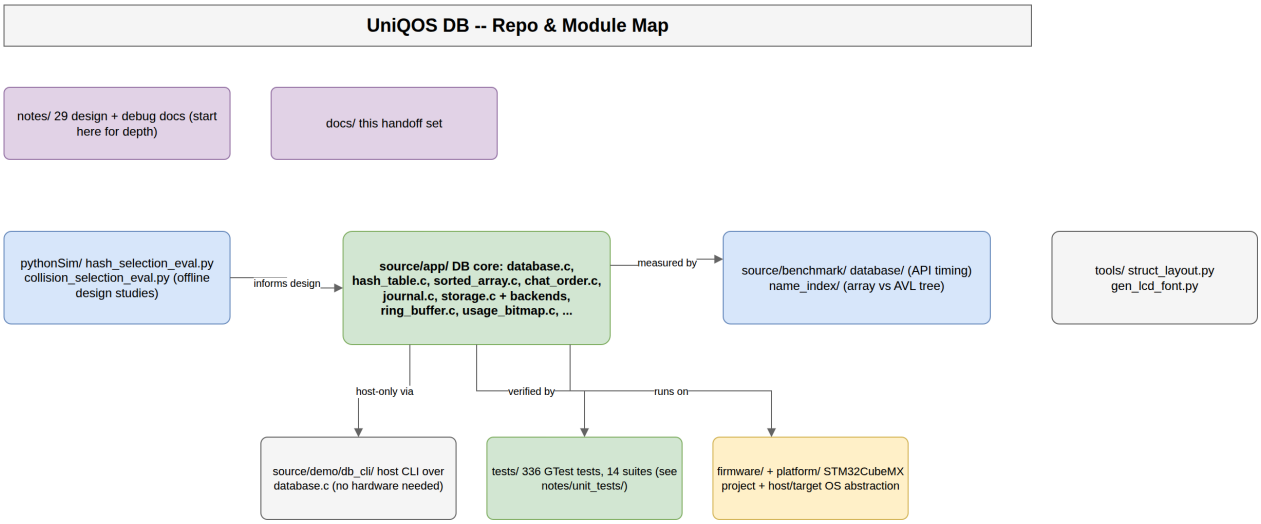


Figure 1: Where each directory fits: design studies inform the DB core, which is verified by the test suite and measured by the benchmarks, and runs on the STM32 firmware (or standalone via the host CLI).

At a glance

Target MCU	STM32H723ZGTx (Cortex-M7), Nucleo-based board
Design capacity	10,000 contacts, 2x that in messages
Hash table	14,293 slots, DJB2 hashing, double hashing + tombstoning, 70% load cap
Sector size	512 B – 6 contacts/sector (81 B each), 2 messages/sector (164 B each)
RAM usage	200.092 KB / 320 KB of RAM_D1 (AXI SRAM) for the hash table + name index + free-list stacks
Test suite	336 GTest tests across 14 suites, run host-side via `ctest`
Build system	CMake; separate `build-host/` (GTest) and `build/` (STM32, Ninja + toolchain file) trees